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Multimodal image fusion: A systematic review

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ABSTRACT

Multimodal image fusion combines information from multiple modalities to generate a composite image containing complementary information. Multimodal image fusion is challenging due to the heterogeneous nature of data, misalignment and nonlinear relationships between input data, or incomplete data during the fusion process. In recent years, several attention mechanisms have been introduced to enhance the performance of deep learning models. However, little literature is available on multimodal image fusion using attention mechanisms. This paper aims to study and analyze the latest deep-learning approaches, including attention mechanisms for multimodal image fusion. As a result of this study, the graphical taxonomy based on the different image modalities, various fusion strategies, fusion levels, and metrics for fusion tasks has been put forth. The focus has been on various Multimodal image fusion frameworks based on deep-learning techniques as their core methodology. This paper also sheds light on the challenges and future research directions in this field, application domains, and benchmark datasets used for multimodal fusion tasks. This paper contributes to the research on Multimodal image fusion and can help researchers select a suitable methodology for their applications.

1. Introduction

Imagine a scenario where a machine learning model can combine information from multiple modalities such as RGB, thermal, and depth images to detect objects and events with higher accuracy and reliability. Multimodal image fusion can create intelligent systems that identify objects in challenging environments and take appropriate actions in real-time. For example, in medical imaging, multimodal image fusion can help diagnose and treat complex diseases by combining information from different imaging modalities such as computerized tomography (CT) and Magnetic resonance imaging (MRI) scans [1,2]. In remote sensing, the fusion of data from sensors such as radar, lidar, and optical cameras can provide a more comprehensive understanding of the Earth's surface [3]. Although multimodal image fusion is still an emerging technology, recent advances in deep learning and other machine learning approaches have shown promising results (Y. Li, Zhao, Lv, and Li 2021). Combining information from different modalities can improve the accuracy and robustness of various computer vision tasks, such as object recognition, image segmentation, and scene understanding.

Soon, multimodal image fusion is expected to play a critical role in various applications, such as autonomous driving, surveillance systems, and robotics [4]. Integrating information from multiple sources allows these systems to operate in complex and dynamic environments, making them more efficient, reliable, and safe. The progress made in multimodal image fusion technology is accelerating toward a new era of intelligent machines that can work seamlessly with humans, enhancing our capabilities and improving our quality of life [5]. Multimodal image fusion has been used in robotics, remote sensing, surveillance, and medical imaging because of its potential application. Multimodal image fusion can provide a more thorough and accurate image of the scene by merging several features of imaging data, which helps with decision-making and improves end-to-end system performance [6]. Although challenges are still to be overcome, such as dealing with varying illumination conditions and sensor noise, the progress made in multimodal image fusion drives us closer to a future where autonomous systems can operate seamlessly and efficiently without human intervention [7]. While the technology for multimodal image fusion is still nascent, the progress made in deep learning and other non-deep learning approaches is promising. Deep learning methods, such as convolutional neural networks (CNNs), generative adversarial networks (GANs), and Transformer [2,8] based approaches have shown great potential for improving the quality and robustness of multimodal image fusion. Non-deep learning approaches, such as sparse representation-based methods and multi-scale and multi-feature decomposition techniques, have also been used successfully in multimodal image fusion tasks [9]. Multimodal image fusion has become a critical research domain [10,11] in various fields, including medical imaging, remote sensing, robotics, and autonomous driving. Multimodal imaging

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systems, such as MRI, CT, and PET, provide complementary information that can be fused to provide a more comprehensive understanding of the scene.

Additionally, multimodal imaging data can improve decision-making and enhance system performance. The need for multimodal image fusion technology is increasing rapidly due to its broad applicability in various domains. For example, in medical imaging, multimodal image fusion can help improve diagnosis and treatment planning by combining information from different imaging modalities [12]. In remote sensing, multimodal image fusion can help enhance the quality of image and accuracy by combining information from different sensors [3,13]. Furthermore, using multimodal image fusion technology in robotics and autonomous driving can improve object detection, scene understanding, and decision-making. By combining information from different sensors, such as cameras and lidar, multimodal image fusion can provide a more accurate and comprehensive understanding of the environment [14]. The increasing demand for multimodal image fusion technology drives research and development efforts in academia and industry (Mohanad G. Yaseen, Mohammad Naeemullah, and Ibarhim Adeb Mansoor 2021). The market for multimodal image fusion technology is expected to proliferate in the coming years due to the increasing demand for improved system performance and the growing availability of imaging data [15].

Over the past few years, multimodal image fusion has become an important research domain in computer vision and related fields. Many research papers have been published on this topic, covering different aspects of multimodal image fusion, such as fusion techniques, performance evaluation, and applications. However, there is still a need for a systematic review of the progress made by state-of-the-art deep learning techniques in multimodal image fusion. Existing review papers have covered various aspects of multimodal image fusion, but this task lacks an in-depth analysis of deep learning approaches using attention mechanisms. Therefore, this survey paper aims to thoroughly analyze the progress made by deep learning techniques in multimodal image fusion, including various fusion architectures, loss functions, and evaluation metrics. This paper proposes a graphical taxonomy, which is a valuable addition, aiding readers in visualizing the relationships between image modalities, fusion strategies, levels of fusion, and evaluation metrics.

The contribution of this survey paper is to provide a systematic review of deep learning approaches for multimodal image fusion and to identify the current state-of-the-art techniques in this field. This review will be a valuable resource for researchers and practitioners working on multimodal image fusion and help guide future research. This is the first-ever systematic survey of multimodal image fusion addressing the fusion architecture using attention mechanisms.

The work is contributed as follows:

- The graphical taxonomy of multimodal image fusion has been outlined systematically, covering all factors.
- A thorough survey of deep learning approaches for multimodal image fusion is done.
- The trade-offs in multimodal image fusion are discussed from the perspective of traditional approaches and different deep-learning approaches.
- Datasets with different modalities have been listed, and the needs and issues of multimodal image fusion are explored.
- The current challenges, application domains, and possible future directions in the domain of deep learning applicable to multimodal image fusion are thoroughly put forth.

The roadmap of the paper is outlined in Fig. 1. The taxonomy of the multimodal image fusion task is outlined in Section 2. The topics of traditional and deep learning methods for multimodal image fusion are covered in Sections 3 and 4. Sections 6 and 7 briefly overview application domains and benchmarked datasets. Section 8 outlines the challenges in research and potential approaches for multimodal image fusion. The paper is concluded in Section 9.

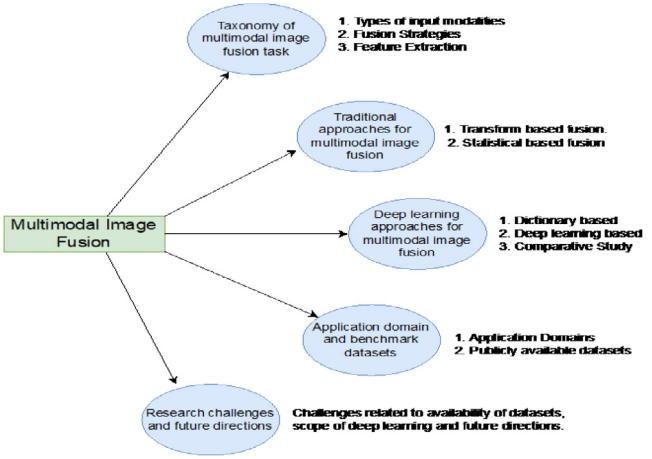

Fig. 1. Roadmap of topics discussed in the paper.

2. Taxonomy of multimodal image fusion task

Multimodal image fusion combines multiple images acquired from different imaging modalities into a single fused image that preserves the most useful and relevant information from each input modality. Multimodal image fusion aims to provide a more comprehensive and accurate representation of the underlying object or scene being imaged than can be obtained from any single modality alone. Fig. 2 depicts the taxonomy of multimodal image fusion tasks. The taxonomy is based on various factors to be considered for the fusion of multimodal images.

2.1. Input modalities

2.1.1. Bi-spectral fusion (two modalities)

Bi-spectral fusion is a type of multimodal image fusion that combines two different modalities, typically visible and infrared images, to provide more comprehensive information about the scene or object being observed. Visible images are sensitive to the scene's colors and brightness, while infrared images are sensitive to the scene's temperature. Combining these two modalities allows bi-spectral fusion to provide more information about the scene, especially in low-light or night-time conditions where visible light is limited. This can be useful in various applications, such as surveillance, navigation, and remote sensing.

For Example, Suppose we want to monitor a forest at night to detect potential wildfires. We can use a visible camera to capture images of the forest during the day and an infrared camera to capture images at night when there is no visible light. However, visible images alone may not detect potential fires, as smoke may not be visible during the day. Fire can be difficult to detect at night using only infrared images.

By combining the visible and infrared images using bi-spectral fusion, we can create a more comprehensive image that combines the color and brightness information from the visible

Image with the temperature information from the infrared image. This can help us to detect potential fires more accurately, as the combination of color, brightness, and temperature information can help to distinguish fires from other heat sources and detect smoke even during the day.

Authors Agrawal and Karar [16] proposed a multi-level fusion algorithm based on curvelet transform for the fusion of registered visible and infrared images of the same scene. Zhou et al. [17] propose a novel enhanced spectral fusion network for hyperspectral image classification. Based on the fusion of different spectral strides, the model is divided into two parts: an optimized multi-scale fused spectral attention module (FsSE) and a 3D convolutional neural network (3D CNN).